Adam Cisneros

Software Engineer | B.S. Computer Science

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Education

California Polytechnic State University – San Luis Obispo Sep 2024 – Jun 2026

Bachelor of Science in Computer Science *GPA: 3.75 | Magna Cum Laude*

- **Awards/Honors:** President's List (2 years), Dean's List (6 quarters)
- **Relevant Coursework:** Data Structures, Algorithms, Operating Systems, Networking, Distributed Computing, Computational Theory, Web Development, Databases, SQL/Data Querying, Artificial Intelligence, Machine Learning, Programming Languages (Compiling, Parsing, etc.), Mobile Development, Game Development

Las Positas Community College Aug 2021 – May 2024

Computer Science *GPA: 3.4 | AI/ML Certificate*

- **Relevant Coursework:** Object Oriented Programming/Design, Artificial Intelligence, Machine Learning

Experience

Software Engineer Intern Jun 2025 – Aug 2025

Amoeba Laboratory LLC – Tracy, CA *amoebalaboratory.com*

- Designed and shipped a full-stack, cloud-based platform end-to-end for real-time food safety operations through gathering and documenting requirements from diverse stakeholders.
- Implemented user authentication and authorization using Firebase Auth.
- Built scalable backend services using optimized data structures, improving response time and system reliability.
- Collaborated with cross-functional teams to define and implement new features based on user feedback.
- Conducted code reviews and provided constructive feedback to peers, fostering a culture of continuous improvement.
- Developed responsive, user-centric UIs and implemented CI/CD pipelines for production-grade deployment.

Leadership Experience

Vice President Jan 2025 – Mar 2026

Cal Poly Game Development Club *cpgd.org*

- Directed multidisciplinary teams through agile development cycles and collaborative software sprints related to game development for 40+ members.
- Organized and led weekly game development workshops for club members.
- Coordinated team projects and facilitated communication between members.
- Represented the club at university events and engaged with potential new members.
- Collaborated with local game studios for guest speaker events and networking opportunities.
- Developed and maintained the club's website and online presence.
- Mentored peers in Git, version control, and design patterns related to game development.

Projects

Distributed P2P File Transfer System | GitHub

- Engineered a custom reliable-transport protocol on top of raw UDP sockets, implementing stop-and-wait ARQ with sequence numbers, timeouts, and retransmission to guarantee in-order, lossless delivery over an inherently unreliable transport layer.
- Implemented full TCP-style congestion control over UDP — slow start, congestion avoidance, and fast retransmit — to dynamically adapt transmission rate to simulated packet loss and network conditions.
- Applied SHA-256 hashing to verify end-to-end data integrity of transferred files, detecting corruption introduced during transmission.
- Built a peer-to-peer file transfer system with a central tracker service for peer discovery, where peers ping the tracker with their available file listings for lookup by other peers.
- **Tech:** Python, UDP Sockets, Custom Protocol Design

Social Content Recommender | GitHub

- Wrote weekly factual analytical, statistical and narrative reports on my iterations analyzing my findings on the user data through SQL queries, graphs, and other data visualizations through my programs and tools like Tableau.

- Performed exploratory data analysis across multi-table user behavior data (views, ratings, metadata), identifying data quality issues (93% null rate in one field, skewed distributions, rounding anomalies) and using summary statistics and Tableau visualizations to surface engagement patterns by genre, region, and cohort.
- Built and iterated a K-nearest-neighbors recommender using weighted user-content similarity, then diagnosed a subtle date-arithmetic bug in a custom calendar system that was distorting recency weighting — fix improved F1 by 0.05 and AUC by 0.05 on held-out evaluation data.
- Engineered features (one-hot encoding, TF-IDF, StandardScaler) and benchmarked their impact on model performance, quantifying trade-offs (e.g., high-dimensional TF-IDF degraded KNN but improved a Random Forest classifier) using ROC-AUC, PR-AUC, and F1 across thresholds.
- Implemented and evaluated four recommendation strategies (global heuristic, collaborative filtering, content-based via XGBoost, and a hybrid blend) using NDCG@k, ROC/PR curves, and F1, explaining performance differences via the cold-start problem and identifying which user segments each model served best.
- Built a churn prediction pipeline (XGBoost) with time-based train/test splitting to prevent temporal leakage, engineered engagement features (watch fraction, unique content viewed, watch time), and tuned the classification threshold via F1-optimized threshold analysis.
- Applied Alternating Least Squares matrix factorization to derive latent user embeddings from a sparse interaction matrix, reduced dimensionality with UMAP, and clustered with K-means (validated via elbow method and silhouette scores) to generate 7 data-driven user personas used to inform churn and recommendation strategy.
- **Tech:** Python, SQL, scikit-learn, Machine Learning, Artificial Intelligence, Data Analysis, Data Visualization

Valley Brews (Simulated Commerce Market) | GitHub

- Designed and implemented a normalized PostgreSQL schema (via SQLAlchemy) supporting 100k+ product variants, replacing hardcoded logic with fully data-driven catalog, inventory, and pricing queries.
- Built a ledgerized (append-only) accounting system for gold, inventory volume, and stock levels, using SUM-based on-the-fly aggregation instead of mutable state — enabling point-in-time historical reconstruction and event-level reconciliation of discrepancies.
- Modeled relational order-management system with carts and $cart_i temstablesandenforcedforeignkeyintegrity, migratingorderstatefromin-memorystructurestodurable, queryabledatabaserecords.Developedasearch/filter/sort/paginationendpointovertransaction$
- Wrote raw SQL (SELECT, UPDATE, INSERT, aggregate/window queries) executed via SQLAlchemy against a live PostgreSQL backend processing simulated transactions on a recurring interval (12 ticks/day).
- Iteratively evolved the schema across 5 versions — from a single-row inventory table to a fully normalized, ledger-backed, audit-verified relational database — while maintaining backward-compatible API contracts
- **Tech:** Python, SQL, scikit-learn, Machine Learning, Artificial Intelligence, Data Analysis, Data Visualization

Game Success Classifier | GitHub

- Collected a custom dataset of 1,200 recent games via the RAWG API (sampled across a 5-year window) to avoid staleness in public game datasets, then used Tableau/Python EDA to uncover a right-skewed, multimodal user-rating distribution and a hidden data-quality bug (API used 0 to encode missing ratings, contaminating the raw signal).
- Found no linear correlation between critic (Metacritic) and user ratings through exploratory analysis, motivating a predictive model to quantify and explain the discrepancy between critic and audience reception.
- Engineered features including TF-IDF-encoded genres/developers/publishers and aggregated historical performance metrics (mean user/critic scores per developer and publisher), using a time-based 80/20 train/test split (sorted by release date) to prevent data leakage.
- Built a multi-output XGBoost regressor to simultaneously predict user rating and critic score, then applied a Gaussian Mixture Model to data-derive quadrant decision boundaries (rather than fixed thresholds), better matching the skewed, multimodal rating distribution.
- Achieved 87% classification accuracy (Macro F1: 0.82) and strong regression fit (R^2 of 0.84 and 0.79) in predicting whether a game would land in high/low user-and-critic-reception quadrants, validated via confusion matrix and MAE/RMSE.
- Diagnosed model limitations through confusion matrix analysis, identifying that prediction accuracy dropped specifically for games where user and critic sentiment diverged which are the exact cases the project set out to explain.
- **Tech:** Python, SQL, Artificial Intelligence, Machine Learning, scikit-learn, Data Analysis, Data Visualization

GamePlan | GitHub

- Built a cross-platform mobile app in Kotlin using Android Studio with Gradle for build configuration and dependency management.
- Integrated Firebase Auth for secure user authentication and Firestore as a scalable NoSQL backend database.

- Leveraged GitHub's REST API to seamlessly pull user account data into the app, reducing manual input and improving onboarding.
- Architected a hierarchical Firestore data model to manage projects, tasks, and ideas across user groups with role-based access and shared workspaces.
- Designed a dynamic, responsive UI using Jetpack Compose and RecyclerView with custom adapters to display and organize user-generated content.
- **Tech:** Kotlin, Android Studio, Gradle, REST API

City-Management/Eco-Simulator Game | GitHub

- Designed and implemented a normalized PostgreSQL schema (biomes, entities, villagers, jobs, buildings, storage) with foreign-key-enforced relationships to model a self-updating simulated ecosystem and village economy.
- Load-tested the database by synthetically generating 1.2M+ villager records via a custom data-generation script, then benchmarked 15+ API endpoints under this scale to identify the slowest-performing operations.
- Used PostgreSQL's EXPLAIN to analyze query plans on the slowest endpoints (a 1.2M-row bulk UPDATE at 10.7s, a 90k-row job-assignment UPDATE at 2.2s), diagnosing sequential-scan bottlenecks and testing targeted indexes (partial index on $\text{job}_id = 0, index on resource_name) to evaluate real performance impact. Determine empirically that indexing did not always help^- for sm$
- Identified and documented concurrency hazards (dirty reads, phantom reads) in the transaction flow through sequence diagrams, then redesigned read/update ordering (batching reads at menu load, deferring writes until transaction confirmation) to eliminate stale-data bugs.
- Built REST API endpoints (auth, admin, ecological simulation, village management, job assignment) backed by SQLAlchemy against a Supabase-hosted PostgreSQL instance.
- **Tech:** Python, SQL, Pygame, Postgres, SQLAlchemy, Vercel, RESTful APIs

Virtual Closet Web App | GitHub

- Built full-stack features for myCloset, a MERN-style closet inventory app, developing both React/Tailwind UI components and backend REST API endpoints consumed by the frontend.
- Implemented JWT-based authentication and authorization (jsonwebtoken, bcrypt password hashing) securing protected API routes and user sessions.
- Designed and built the outfit creator feature, including logic for randomly generating outfit combinations from a user's saved closet inventory.
- Developed the closet inventory UI in React, allowing users to browse, add, and manage clothing items in one interface using React Router for client-side navigation.
- Wrote Jest unit tests for core components (e.g., item upload form) and contributed to a GitHub Actions CI/CD pipeline running lint and build checks on every push.
- Collaborated in a 5-person Agile team using GitHub Issues/PRs for sprint planning, code review, and iterative feature delivery, following a documented sequence-diagram-driven API design process.
- **Tech:** React, MongoDB, TailwindCSS, RESTful APIs, JavaScript, Azure, CI/CD

Foiled! | itch.io Page

- Leading development of a tournament-style fencing game featuring a diverse roster of opponents with distinct playstyles.
- Building in Godot with advanced AI and a machine learning model that dynamically adapts enemy behavior to counter the player's fighting style.
- **Tech:** Godot, Artificial Intelligence (AI), Machine Learning (ML), GDScript

Build By Bubble (GGJ 2025) | GitHub | itch.io Page | GGJ Page

- Shipped a complete game in Unity within 48 hours for the 2025 Global Game Jam under the theme "Bubble".
- Designed a puzzle-platformer where players use bubbles to construct platforms and navigate through levels.
- Collaborated with a partner, owning all gameplay programming from mechanics to player controls.
- **Tech:** Unity, C#, Gameplay Programming

Aelar: The Celestial Curse (GGJ 2026) | GitHub | itch.io Page | GGJ Page

- Shipped a bullet hell game in Godot within 48 hours for the 2026 Global Game Jam under the theme "Mask".
- Features wave-based enemy combat with a loot and equip system that lets players steal and wield defeated bosses' powers.
- Led gameplay programming within a team, including full implementation of the inventory and equipment system.
- **Tech:** Godot, GDScript, GamePlay Programming

There Can Only Be One | GitHub

- Developed a time-travel stealth-action game in Unity where players navigate a bank heist using time manipulation to outsmart guards and solve puzzles.
- Features a paradox mechanic in which past selves accumulate over time and actively work against the player to prevent timeline alterations.
- Designed around high-stakes decision-making, where every action carries consequences that ripple across the timeline.
- Led gameplay and AI programming for guard NPCs and past-self adversaries within a collaborative team.
- **Tech:** Unity, C#, Artificial Intelligence (AI), Gameplay Programming

Technologies

Languages: Python, C++, C, C#, JavaScript, HTML, CSS, SQL

Systems & Tools: REST APIs, DBMS, Cloud Platforms (Azure, Vercel), Database Systems (SQL & NoSQL), CI/CD, Git, GitHub, React, MongoDB, Firebase/Firestore, Supabase, scikit-learn, StableBaselines3, *nix (Linux & Unix-like OS), Tableau, Excel

Concepts: Full-Stack Development (Backend & Frontend Design), Scalable Systems, Distributed Systems, Cloud Deployment, UI/UX, Artificial Intelligence & Machine Learning, Data Structures & Algorithms, Agile Collaboration & Development, Data Analysis, Data Visualization, Statistical Modeling, Authentication, Data Flow, Unit Testing, Software Development Life Cycle (SDLC), Networking (UDP, TCP/IP, etc.)